1(a)	$(1-4x)^7$	
	Correct form for term 3 or term 4: Either $\frac{7\times6}{2}(...x)^2$ **or** $\frac{7\times6\times5}{6}(...x)^3$ o.e.	M1
	$\frac{7\times6}{2}(-4x)^2$ **or** $\frac{7\times6\times5}{6}(-4x)^3$ o.e.	A1
	$=\underline{1-28x}+336x^2-2240x^3$	<u>B1</u>, A1
		(4)
(b)	Sets $"336"\times5-"28"k=1316$	M1
	$(k=)\,13$	A1
		(2)
		(6 marks)

2(a)	Midpoint of $(-2,5)$ and $(4,15)$ is $\left(\frac{-2+4}{2},\frac{5+15}{2}\right)=(1,10)$	B1
	Attempts radius ^{2} or diameter ^{2}: e.g. $D^2=(4--2)^2+(15-5)^2=136$	M1
	$\text{Radius}^2 = 34$	A1
	$(x-1)^2+(y-10)^2=34$	M1, A1
		(5)
(b)	$\left(1,10-\sqrt{34}\right)$	B1 , dB1
		(2)
		(7 marks)

3	$\int_1^4 3x+\frac{16}{x^2}-8\,\mathrm{d}x=\frac{3}{2}x^2-\frac{16}{x}-8x \quad (+c)$ e.g. Area $=\frac{3}{2}(5+11)-\left[\frac{3}{2}x^2-\frac{16}{x}-8x\right]_1^4=24-\left(-12+\frac{45}{2}\right)=\frac{27}{2}$	M1, A1, A1 dM1, A1 **(5)** **(5 marks)**

	$f(x) = ax^3 + bx^2 + 5x - 3$	
4(a)	Sets $f(-3) = 0 \rightarrow \ a \times (-3)^3 + b \times (-3)^2 + 5 \times (-3) - 3 = 0$	M1
	$\Rightarrow -27a + 9b = 18 \Rightarrow \ b = 3a + 2$ *	A1*
		(2)
(b)	Sets $f\left(\frac{1}{2}\right) = \frac{7}{4} \rightarrow \ a \times \left(\frac{1}{2}\right)^3 + b \times \left(\frac{1}{2}\right)^2 + 5 \times \left(\frac{1}{2}\right) - 3 = \frac{7}{4}$	M1
	$a + 2b = 18$	A1
	Solves $b = 3a + 2$ with their $a + 2b = 18 \Rightarrow a = ..., b = ...$	M1
	$a = 2, b = 8$	A1
		(4)
(c)	e.g. $\frac{2x^3 + 8x^2 + 5x - 3}{x - 2} = \left(...x^2 + ...x + ...\right) + \text{Remainder}$	M1
	(Quotient =) $\quad 2x^2 + 12x + 29$	A1
	Remainder = 55	B1
		(3)
		(9 marks)

5(a)	Identifies $h = 1.5$	B1
	Area $= \frac{1.5}{2}\left\{12 + 0.023 + 2\left(4.243 + 1.5 + 0.530 + 0.188 + 0.066\right)\right\}$	M1
	$= 18.8$	A1
		(3)
(b) (i)	$\int_{-2}^{7} 3\left(\frac{1}{2}\right)^{x+2} \mathrm{d}x = \frac{1}{4}\int_{-2}^{7} 3\left(\frac{1}{2}\right)^{x} \mathrm{d}x = \frac{1}{4} \times \text{"18.8"} = 4.7$	B1ft
(ii)	$\int_{-2}^{7} \left(2^{-x} + 2x\right) \mathrm{d}x = \frac{1}{3} \times \text{"18.8"} + \left[x^2\right]_{-2}^{7} = 51.3$	M1, A1ft
		(3)
		(6 marks)

6. (i)

$$2\log_2(4-x) = 3 + \log_2\left(\frac{x+11}{2}\right)$$

One correct law applied – examples (not exhaustive):

$2\log_2(4-x) = \log_2(4-x)^2$, $\log_2\left(\frac{x+11}{2}\right) = \log_2(x+11) - \log_2 2$

$3 = \log_2 8$ — B1

Correctly combines two original terms e.g. $2\log_2(4-x) = \log_2(4x+44)$ — M1

Correct equation not involving logs $(4-x)^2 = 4x+44$ — A1

$x^2 - 12x - 28 = 0 \Rightarrow (x-14)(x+2) = 0 \Rightarrow x = ...$ — dM1

$\Rightarrow x = -2$ only — A1

(5)

(ii)

Combines $y = 6\times 3^x$ and $y = 3^{2x+1} \Rightarrow 3^{2x+1} = 6\times 3^x$ — M1

$\Rightarrow 3^x \times 3 = 6$

$\Rightarrow 3^x = 2 \Rightarrow x = \log_3 2$ — dM1, A1

$y = 6\times 3^{\log_3 "2"} = 6\times "2"$ — ddM1

coordinates of P are: $\left(\log_3 2,\ 12\right)$ — A1

(5)

(10 marks)

7. (i)	$10+8+6.4+...$	
(a)	Attempts $S_\infty = \dfrac{a}{1-r}$ with $a=10, r=\dfrac{8}{10}$	M1
	$S_\infty = 50$	A1
		(2)
(b)	Sets $10\times 0.8^{k-1} < 0.0005 \Rightarrow 0.8^{k-1} < 0.00005$	M1
	e.g. $\Rightarrow k-1 > \dfrac{\log(0.00005)}{\log(0.8)}$ or e.g. $k-1 > \log_{0.8} 0.00005$	M1
	$\Rightarrow k-1 > 44.38 \Rightarrow (k=)\ 46$	A1
		(3)
(ii)	Sets $850+(n-1)\times -7 = 0 \Rightarrow n = 122.4$	M1
	States or implies that $n=122$	A1
	$S = \dfrac{122}{2}\{2\times 850 + 121\times -7\} = 52033$	dM1, A1
		(4)
		(9 marks)

8 (i)	$3\tan\left(2x+\frac{\pi}{5}\right)=\sqrt{3}\Rightarrow 2x+\frac{\pi}{5}=\arctan\left(\frac{\sqrt{3}}{3}\right)$	M1
	$\Rightarrow 2x+\frac{\pi}{5}=\frac{\pi}{6},\frac{7\pi}{6},\frac{13\pi}{6}\Rightarrow x=...$	dM1
	$\Rightarrow x=\frac{29\pi}{60},\frac{59\pi}{60}$	A1
		(3)
(ii)	$5\sin\theta\tan\theta=\cos\theta+4$	
	$\tan\theta=\frac{\sin\theta}{\cos\theta}\rightarrow \quad 5\sin\theta\times\frac{\sin\theta}{\cos\theta}=\cos\theta+4$	M1
	$\Rightarrow 5\left(1-\cos^2\theta\right)=\cos^2\theta+4\cos\theta$	dM1
	$\Rightarrow 6\cos^2\theta+4\cos\theta-5=0$	A1
	$\Rightarrow \cos\theta=\frac{-4\pm\sqrt{16+120}}{12}$ **(**$=0.63849...$**)** $\Rightarrow\theta=$	ddM1
	$\Rightarrow \theta=50.3°, 309.7°$	A1
		(5)
		(8 marks)

9 (a)	Attempts $12 = 3x^2 \times L \Rightarrow L = \frac{4}{x^2}$ or $Lx = \frac{4}{x}$	M1, A1
	$S = 6x^2 + 5xL = 6x^2 + \frac{20}{x}$	dM1, A1
		(4)
(b)	$\frac{dS}{dx} = 12x - \frac{20}{x^2}$	M1, A1ft
	$\frac{dS}{dx} = 0 \Rightarrow x^3 = \frac{5}{3} \Rightarrow x = 1.18563...$	dM1
	$S\big\|_{x=1.18563} = 6 \times "1.18563"^2 + \frac{20}{"1.18563"} = 25.3\,(\text{m}^2)$	ddM1, A1
		(5)
(c)	$\left(\frac{d^2S}{dx^2} =\right) 12 + \frac{40}{x^3}$	M1
	Justifies $\frac{d^2S}{dx^2} > 0$ (at $x = 1.18563...$), hence minimum *	A1*
		(2)
		(11 marks)

10	Sets $m = 3p+1$ or $m = 3p+2$ ($p > 0$) o.e. and attempts $m^2 + 3m + 2$	M1
	E.g. $m^2 + 3m + 2 = (3p+1)^2 + 3(3p+1) + 2 = 9p^2 + 15p + 6 = 3(3p^2 + 5p + 2)$	A1
	Sets $m = 3p+1$ and $m = 3p+2$ ($p > 0$) and attempts $m^2 + 3m + 2$	dM1
	E.g. $m^2 + 3m + 2 = (3p+2)^2 + 3(3p+2) + 2 = 9p^2 + 21p + 12 = 3(3p^2 + 7p + 4)$	
	And gives a minimal conclusion *	A1* **(4 marks)**